

Neural Array Projection Oscillation Tomography

A Ten-Stage Genealogy, Frozen Neuropixels Protocol, and Synthetic Failure Calibration

Author: Micah Blumberg

Self Aware Networks Research Institute

<https://selfawarenetworks.com>

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Abstract

Neural Array Projection Oscillation Tomography (NAPOT) proposes that distributed neural populations receive partial measurements, transform them through state-dependent temporal structure, project consequences, and recurrently update an action-relevant estimate of body and world. Draft 1 recovered ten developmental stages and converted their joined mechanism into a falsifiable latent-state framework. Draft 2 froze a multisite Neuropixels benchmark with eight capacity-matched model families, eleven nulls and ablations, mouse-isolated splits, sequential neural and action outcomes, and explicit failure conditions. Draft 3 began implementation without opening biological outcomes: a deterministic five-seed fixture exercised all eight information-admission contracts, all eleven destruction routes, mouse-disjoint splits, leakage guards, and causal phase and phase-wave-differential operators. Draft 4 added a capacity-matched recurrent architecture suite under one 2,146-parameter contract. Draft 5 triangulated the written ladder against four public 2022 explainer transcripts: 1,534 of 1,534 deterministic segments were reviewed, yielding 89 bounded authorial atoms. Draft 6 freezes the videos' live metadata and archive custody and calibrates the previously inconclusive synthetic action endpoint across nine conditions and 20 independent seeds per condition. The original five-seed result reproduces exactly. The zero-signal negative control wins only 6/20 seeds and fails the descriptive stability rule. At the baseline generator setting, M7 wins 15/20 seeds with a median advantage of -0.00570 bits per trial, but its 90th-percentile delta is +0.00585, so it also fails the stability rule. None of the nine conditions passes. The result identifies an architecture or endpoint limitation rather than converting synthetic evidence into a positive claim. Public recordings strengthen chronology and source fidelity, not biology. Synthetic results are engineering evidence about recoverability, causality, model routing, and failure behavior. They are not evidence that brains implement NAPOT, that phase is generally superior to rate or power, or that a decoded state is conscious. The operators have not been bound to Allen schemas, the biological file manifest remains empty, and no Allen file, test identity, or outcome has been opened. The present paper closes at the genealogy, protocol, executable-operator, recurrent-model, and synthetic failure-calibration boundary. A later Allen execution remains a separately preregistered biological study; it is not an unreported positive result or missing evidence for this paper's narrower methods claim.

1. The problem before the acronym

A nervous system does not receive a complete scene in one place. Different receptors and neural populations are sensitive to different features, locations, time scales, body states, and possible actions. Their signals are delayed, noisy, overlapping, and changed by the organism's own movement. Useful behavior nevertheless requires a current estimate: what is present, where it is relative to the body, what is changing, what matters, and what will happen if the organism acts.

It is tempting to answer this problem with a picture metaphor. Many signals converge, the brain creates an image, and a self looks at it. That merely moves the explanatory burden. Who reads the image? How does that reader select an action? How is the image corrected after the action changes the sensory input? A mechanistic account must close the loop without installing a second observer inside the first.

NAPOT begins with a different intuition. A population can receive structured input, transform its state, project consequences to other populations, and receive the changed result. A useful estimate need not be displayed to a viewer. It can exist in the selective causal organization of the system: which next states become possible, which errors are reduced, and which actions are chosen. In this paper, **rendering** means the recurrent construction and updating of such an action-relevant latent state. It does not mean a literal screen.

The term **Neural Array Projection Oscillation Tomography** is earned in four steps:

1. *neural array* refers to distributed populations with partially distinct observations and outputs;
2. *projection* refers to learned, directed consequences of one population for another, not only an anatomical axon bundle;
3. *oscillation* refers to measured temporal and phase structure when that structure adds predictive information; and
4. *tomography* is a conditional inverse-problem claim, not a decorative synonym for integration.

1.1 The protocol question before the mechanism

The public genealogy begins before the NAPOT acronym. In the April 2017 *Neural Lace Podcast*, Blumberg asked whether communication in the brain was more like TCP, with connection and feedback, or UDP, with connectionless transmission. The discussion proposed that a biological network might alternate between such modes and deliberately named the hybrid **ATP: Alternating Transmission Protocol**. The acronym was chosen because ATP also names adenosine triphosphate, the cellular energy currency, making communication and metabolic expenditure two sides of the same biological-network question [12].

This is a problem-formulation ancestor, not evidence that neurons literally implement Internet protocols. PWD and NAPOT later offer a more specific candidate answer: relative phase, magnitude, timing, feedback, and projection could define measurable mode changes in recurrent neural communication. The 2017 source therefore establishes continuity of the question while leaving the later mechanism and its biological validation open.

2. What the ten revisions changed

The recovered source record supports ten developmental stages. Commit dates, in-file development dates, and scientific validation are different facts. Some sources are Micah-authored notes; some are attributed transcripts; some are AI-assisted dialogues in which Micah's prompts preserve a proposal but the model's prose is not scientific authority.

2.0 Public spoken-source triangulation

The written ladder is now independently routed through four public 2022 video transcripts. A deterministic index divides the raw bodies into 1,534 stable segments: 28 from July 28, 135 from July 30, 492 from July 31, and 879 from September 18. Every segment has been reviewed, and 89 bounded authorial atoms have been recorded. The atom count is a source-coverage result, not 89 empirical findings or priority verdicts.

Public source layer	Stable locator	NAPOT content preserved	Boundary
July 28, 2022	SAN-YT-2022-07-28, 28/28 segments	oscillations bind array-to-array models; synaptic memories can be magnified toward network scale	authored mechanism statement, not biological validation
July 30, 2022	SAN-YT-2022-07-30, 135/135 segments	receive-transform-project architecture; phase-sensitive variable output; memory scaling; named Tse-Buzsaki-Strogatz Choice synthesis	donor atoms retain their original credit; universal output claims remain hypotheses
July 31, 2022	SAN-YT-2022-07-31, 492/492 segments and 31 atoms	projection-slice inspiration, distributed observer, synapse-to-whole-brain scaling, point / volume rendering, semantic segmentation, and sensorimotor closure	computational analogies do not establish algorithmic identity in the brain
September 18, 2022	SAN-YT-2022-09-18, 879/879 segments and 47 atoms	the denser fifth-revision oscillation, inhibition, phase, rendering, memory, and action account	exact medical claims require independent source-by-source review

This triangulation changes the genealogy in one important way. NAPOT was not reconstructed solely from later notes or AI dialogues: a substantial joined mechanism was publicly explained in 2022. It does not change the empirical status of any mechanism. A governed live-platform and archive receipt now freezes all four video IDs, titles, exact upload times, shared channel identity, durations, live watch-page and oEmbed response hashes, and available Wayback CDX captures. The receipt has SHA-256 18E88571A9DBD72916DE745BE3200BB6159A6A8946EB9C718D589659520EF9F2. Corrected transcripts remain hash-linked derivatives of the raw files; caption-track retrieval is optional enrichment rather than an unresolved custody claim.

2.0.1 Recorded-source authority

The source archive contains multiple generations of transcription, and the older versions can contain serious recognition errors. This paper therefore uses the original public audio or video plus native service metadata as the authority for date and wording. Clearer 2026 Whisper retranscriptions support semantic navigation; an older transcript can locate a candidate passage but cannot, by itself, support a quotation, a priority-bearing phrase, or a mechanistic distinction. The 2017 ATP discussion is paraphrased here and routed to the public recording rather than presented as a transcript-exact quotation. `SOURCE-MAP.md` preserves the full authority hierarchy.

2.1 From arrays to finer resolution

NAPOT 1 supplies the durable core: neural arrays receive changing patterns, transform them, project new patterns, and participate in recurrent oscillatory feedback. The original tomography and rendering language asked how many partial neural views could contribute to one usable state and how that state could again influence many populations.

NAPOT 2 identifies a resolution problem. Treating a neuron as one scalar output loses branch-specific nonlinearities, synapse-specific modulation, and the learned morphology that determines which combinations matter. The historical basal-dendrite-as-input and apical-dendrite-as-output metaphors helped pose the receive / project distinction, but they cannot be universalized across cell types or cortical circuits.

The retained question is measurable: does a branch- or compartment-aware model explain held-out neural and behavioral data better than a whole-cell scalar model?

2.2 From positive events to relative departures

NAPOT 3 asks whether departures below a tonic reference can carry information, not only bursts above it. This is a valuable correction to a spike-only story. Its early terms *anti-spike* and *negative soliton* are not treated as established mechanisms. Reduced firing, active inhibition, silence, slow cortical potentials, hyperpolarization, and a propagating negative wave are different observables. A modern test must define which variable departs from which reference and how that departure predicts a later state.

NAPOT 4 combines finer resolution with changing state. A structural network persists over longer periods, while inhibition, excitability, oscillatory context, and recent activity select a faster effective subnetwork. This moves the proposal away from one fixed wiring diagram. It predicts that the functional path carrying a distinction should change with context even when anatomy is held constant.

2.3 From local pattern to multiscale sensorimotor loop

NAPOT 5 expands the account across cell, assembly, area, and body scales. Memory, prediction, reference frames, neural tuning, phase, magnitude, and changing functional connectivity enter as parts of one recurrent process. The scientific repair is to avoid assigning semantic roles to fixed frequency bands in advance. Phase, amplitude, duration, propagation, coupling, and state must be measured and ablated separately.

NAPOT 6 adds the **inceptive field**, proposed as an action-side complement to a receptive field. A receptive-field model asks which prior inputs predict a unit's response. An inceptive-field model asks which downstream neural, bodily, and sensory consequences are predicted by the unit or population's output. This closes the loop from sensing to doing. It also introduces the phase-wave differential as a candidate relative event: a structured change in phase, frequency, magnitude, duration, or propagation compared with a receiver-relevant tonic state. The term does not license the assumption that every difference is meaningful; predictive and intervention tests must select which differences matter.

2.4 A historical fork and a return to operational neuroscience

NAPOT 7 branches into quantum biology and the physics questions that later contributed to QGTCD. That branch belongs in the intellectual genealogy, but protein-spin memory, weak-field control, or quantum neural computation are not used as premises in this paper. Removing them from the empirical core lowers the claim burden and permits the neural reconstruction hypothesis to stand or fall on ordinary measurements.

NAPOT 8 introduces **entification**: physically distinct oscillators may participate in one functionally integrated sensor and actor. The operational content is not that each cell is a person or that synchrony creates a subject. It is that disturbance to one selected component should have predictable, selective consequences for the shared state, action, and recovery of the larger system.

NAPOT 9 returns to three-dimensional sensory reconstruction. It asks whether distributed excitatory and inhibitory activity constrains a body-centered state containing spatial relations. The retained question is an inverse problem. The paper does not assume that the solution is a picture or that excitation and inhibition always encode perception and memory in fixed, opposed roles.

NAPOT 10 connects reconstruction to neural manifolds, continuous approximation, dimensional transformation, choices, and behavior. This is the point at which NAPOT must confront strong standard models. Low-dimensional latent geometry is common in neural analysis. NAPOT adds scientific value

only if its spatial projection, directional propagation, multiscale phase, dynamic gating, and action-closure constraints explain held-out data better than a flexible latent model without those commitments.

No stable NAPOT 11 is currently admitted. One recovered prompt requests stages 11-13, but its generated answer conflicts with the explicit later assignments for stages 8-10. A new revision requires an authored scientific delta, not numbering momentum.

3. Operational core

Let s_t denote a latent state at time t . It may include external stimulus variables, body configuration, task context, intended action, and unobserved neural state. Array a , for $a = 1, \dots, A$, produces a measurement:

$$y_{a,t} = O_a(s_t) + \epsilon_{a,t}. \quad (1)$$

The observation operator O_a is partial and may be nonlinear. The noise term includes measurement noise and unmodeled biological variation. A NAPOT model does not assume that any array sees the whole state.

Each array constructs a directed projection using its measured history and a context or gating state $q_{a,t}$:

$$p_{a,t} = P_a(y_{a,\leq t}, q_{a,t}, x_a), \quad (2)$$

where x_a contains spatial or anatomical coordinates when known. The projection can include rate, event timing, phase, power, duration, propagation direction, and coupling features. These features must be nested and ablated; they are not all presumed necessary.

A recurrent reconstruction operator combines distributed projections, actions, and prior state:

$$\hat{s}_t = R_\theta(p_{1:A,\leq t}, u_{1:t-1}, \hat{s}_{t-1}). \quad (3)$$

Here u_t is an action or experimentally controlled intervention. The reconstruction is useful only if it predicts variables that were not used to fit it. A sensory-only decoder is therefore insufficient. The candidate model also predicts action and reafferent consequence:

$$(\hat{u}_t, \hat{y}_{t+1}) = G_\theta(\hat{s}_t, c_t), \quad (4)$$

where c_t is current task context.

One training objective is:

$$\mathcal{L} = \mathcal{L}_{state} + \lambda_u \mathcal{L}_{action} + \lambda_y \mathcal{L}_{next\ observation} + \lambda_c \mathcal{R}_{capacity} + \lambda_s \mathcal{R}_{stability}. \quad (5)$$

The capacity and stability penalties prevent a more complicated NAPOT model from winning merely because it has more parameters or unstable flexibility. The primary scientific comparison must use held-out performance under matched resource envelopes.

4. When tomography is more than a metaphor

Tomography normally recovers hidden structure from multiple projections under a known or estimable geometry. Neural data present a harder problem. The observation operators may be incomplete, nonlinear, state-dependent, and changed by movement. Different latent states can produce nearly identical recordings. A smooth decoder can fit such data while recovering the wrong hidden variables.

The paper therefore uses a five-part tomography gate:

1. **Projection definition:** the measured views and their geometry or functional operators are specified.
2. **Observability:** the measurements contain enough independent information to constrain the proposed latent variables.
3. **Identifiability:** alternative latent states or parameterizations are characterized rather than silently treated as unique.
4. **Held-out reconstruction:** recovery succeeds across new trials, sessions, subjects, stimuli, or behaviors.
5. **Intervention generalization:** the model predicts consequences when the system is perturbed, not only correlations in familiar data.

For a locally linearized dynamical model $s_{t+1} = Fs_t + Bu_t$, $y_t = Hs_t + \epsilon_t$, a basic observability check uses

$$\mathcal{O}_K = \begin{bmatrix} H \\ HF \\ \vdots \\ HF^{K-1} \end{bmatrix}. \quad (6)$$

Full column rank is sufficient for observability in the ideal linear case. Real neural systems require local sensitivity, posterior uncertainty, simulation-based recovery, and perturbation analysis. Failure of those checks does not prove that the brain lacks a useful state. It means the experiment cannot justify a tomographic interpretation.

5. Frozen comparator registry

NAPOT is evaluated as a nested model family, not against a single weak baseline. Every trainable model uses latent dimension 32, the same train / validation / test assignments, the same five random seeds, the same hyperparameter grid, and a trainable-parameter count within five percent of the reference model. Where architectures make exact matching impossible, a width adjustment is selected from training data before validation and then frozen.

ID	Model	Information admitted	Question answered
M0	Stimulus-behavior null	Stimulus and movement covariates only	How much can be predicted without neural state?
M1	Rate-only recurrent	Spike counts	Does ordinary population rate suffice?
M2	Rate plus power	Spike counts and spectral amplitude	Does oscillatory magnitude help without phase?
M3	Undirected connectivity	M2 plus symmetric coupling summaries	Does connectivity help without propagation direction?
M4	Phase-aware nondirectional	M2 plus circular phase features; coordinates removed	Does phase help without spatial direction?

ID	Model	Information admitted	Question answered
M5	Flexible nonlinear state-space	Learned latent dynamics without NAPOT structural constraints	Does a strong generic manifold model perform as well?
M6	Spatial reconstruction without action closure	Coordinates and directed phase; no action or reafference head	Does spatial reconstruction suffice without sensorimotor recurrence?
M7	Full NAPOT candidate	Coordinates, directed phase, dynamic gating, multiscale state, action and reafference heads	Does the joined NAPOT constraint set add held-out value?

Let P_m be the trainable parameter count for model m and P_5 the count for the flexible state-space comparator. Capacity matching requires

$$\frac{|P_m - P_5|}{P_5} \leq 0.05, \quad m = 1, \dots, 7. \quad (7)$$

All models use Adam, at most 200 epochs, early-stopping patience 20, learning rates $\{10^{-4}, 3 \times 10^{-4}, 10^{-3}\}$, dropout $\{0, 0.1, 0.3\}$, and frozen seeds 20260718-20260722. Validation expected log predictive density selects hyperparameters. The final test set is not used for architecture, normalization, feature, threshold, or stopping decisions.

6. Frozen benchmark protocol

The machine-readable contract is protocol/benchmark-config.json. It is the authority when prose and implementation diverge. Draft 2 freezes data identities, inclusion rules, operators, partitions, models, outcomes, nulls, and statistics before any session file is downloaded or any outcome is opened.

6.1 Dataset identities and roles

Primary: Allen Visual Behavior Neuropixels. Freeze visual-behavior-neuropixels_project_manifest_v0.5.0.json. Eligible sessions must be non-abnormal, contain LFP and complete trial / stimulus tables, and include at least one VISp probe plus two additional visual-area probe-by-structure groups. The confirmatory trial set consists of image-change trials with valid hit or miss outcomes. Catch and false-alarm trials are secondary. The dataset’s paired familiar and novel sessions permit within-program generalization while retaining mouse-level independence [6].

External validation: Allen Visual Coding Neuropixels. Freeze the EcephysProjectCache manifest schema at version 0.2.1 and create an exact file checksum manifest before analysis. Eligible functional_connectivity sessions must contain LFP, Gabor receptive-field presentations, VISp, and two additional visual-area groups. Gabor location is the external reconstruction target; natural- scene identity is secondary. The packaged data include spikes, LFP, visual timing, running, pupil variables, and experiment metadata [5,7].

These datasets were chosen because electrophysiology resolves spike timing and LFP phase. Allen two-photon calcium data may be used later for slow spatial-state replication, but it cannot validate high-frequency phase or propagation claims at its native sampling rate.

6.2 Inclusion, preprocessing, and artifact rules

Use the Allen default unit filters without outcome-dependent relaxation: presence ratio at least 0.95, inter-spike-interval violations at most 0.5, and amplitude cutoff at most 0.1. Preserve the provider’s packaged LFP referencing. Exclude a channel only if more than five percent of samples are nonfinite or required metadata are unresolved. Exclude preregistered segments containing saturation or absolute deviation above eight median absolute deviations, where location and scale are fitted on the training partition. Save every exclusion mask.

Filter LFP from 4 to 150 Hz, remove 60 and 120 Hz line components, and resample after antialiasing to 500 Hz. The spectral grid is {4, 6, 9, 13, 19, 28, 41, 60, 88, 128} Hz. These are measurement coordinates, not semantic bands. No frequency is permanently declared tonic, sensory, cognitive, or motor. Common-median rereferencing, 2-100 Hz bandwidth, and six- or ten-MAD artifact thresholds are sensitivity analyses, not researcher choices made after test inspection.

6.3 Causal phase, PWD, projection, and inceptive-field operators

For channel c , center frequency f , and signal x_c , compute a past-only complex feature

$$z_{cft} = \sum_{k=0}^{K_f} h_f(k) x_{c,t-k}, \quad (8)$$

where h_f is a causal complex filter frozen before testing. Its argument supplies phase and its log magnitude supplies amplitude. No future sample may influence a predictor. The receiver-relative edge phase is

$$\delta_{ijft} = \text{Arg}(z_{jft} z_{ift}^*). \quad (9)$$

For each receiver and frequency, fit a past-only exponentially weighted circular phase and robust amplitude baseline using training data. The operational **phase-wave differential** is not one scalar or one fixed band. It is the vector

$$d_{ijft} = [\text{wrap}(\delta_{ijft} - \bar{\delta}_{ijft}^{\text{past}}), \log |z_{jft}| - \bar{a}_{jft}^{\text{past}}, \ell_{ijft}, v_{ijft}], \quad (10)$$

where e_{ll} is event duration and v is the coordinate-aware lagged propagation estimate. A PWD is retained only when it adds held-out prediction beyond rate and power. The tonic state is the receiver-relative past baseline; a phasic event is a transient informative departure. Either can occur at different frequencies in different states.

An **array** is a probe-by-structure group with registered channel coordinates. Directed projections are training-estimated lagged phase-gradient and spike-transfer features restricted to anatomy-compatible edges. The reconstruction operator is a causal graph-recurrent state estimator. An **inceptive field** is operationally the incremental prediction of later action and reafferent observation attributable to an emitting node or array state after receptive inputs and covariates are included. It is not assumed from anatomy or visual metaphor.

6.4 Leakage-safe partitions

The outer independent unit is the mouse. Canonical mouse IDs are assigned 60/20/20 percent to development, validation, and test using SHA-256 with frozen salt NAPOT-D2-20260718, with only the minimum

stratification needed to preserve primary outcome classes. Every session from one mouse, including familiar and novel sessions, remains in one partition. Within development data, contiguous trial blocks are used and a two-second guard prevents adjacent samples from crossing a split.

Normalization, artifact thresholds, edge selection, baseline states, filters that require fitted parameters, model widths, and hyperparameters are estimated without test access. A secondary stimulus-generalization analysis holds out 20 percent of image identities or Gabor locations within the training-defined strata. The exact mouse and session assignment file is hashed before any test outcome is opened.

6.5 Confirmatory outcomes and multiplicity

The protocol uses fixed sequential gatekeeping.

1. **Gate 1 - future neural prediction.** Over the next 250 ms, M7 must improve held-out expected log predictive density over the best preregistered M1-M6 comparator by more than 0.005 bits per spike. The lower 95 percent mouse-bootstrap bound must exceed the threshold.
2. **Gate 2 - action prediction.** Only if Gate 1 succeeds, M7 must improve hit-versus-miss log predictive density by more than 0.01 bits per trial using neural features from -500 to +150 ms around image change, before the task response window. The lower 95 percent mouse-bootstrap bound must exceed the threshold.

The independent unit is the mouse. Confidence intervals use 10,000 bootstrap resamples over mice with trials nested within mouse. A two-sided mouse-level sign-flip test is exact when feasible and otherwise uses 10,000 permutations. Holm correction applies within each secondary outcome family. Report effect sizes, intervals, every mouse, all five seeds, calibration, and negative results.

Secondary outcomes are reaction-time concordance, calibration error, novel-versus-familiar transfer, external Gabor-location log loss, next-observation error, cross-session generalization, and probe / structure dropout degradation. No secondary result rescues failure of Gate 1.

6.6 Mandatory nulls and ablations

Run every comparison with phase randomization preserving per-channel power, coordinate permutation within structure, propagation-direction reversal, temporal block shuffle, amplitude-matched phase surrogates, movement and pupil covariates added and removed, dynamic-gating ablation, action-head ablation, refference-head ablation, probe and structure dropout, and label permutation. Null transformations are generated independently within each partition. They never mix samples across mouse, session, trial, or split boundaries.

A claimed phase increment fails if it survives power-preserving phase destruction. A directional increment fails if it survives coordinate permutation or direction reversal. A sensorimotor claim fails if it disappears after movement and pupil control or if action / refference heads add no held-out value.

6.7 Observability and simulation recovery

Before fitting biological outcomes, generate synthetic systems with known latent state, projection operators, phase propagation, movement confounds, and intervention effects. The implementation must recover known state and reject nonidentifiable variants under the same splits and metrics. For each biological model, report local sensitivity rank, posterior or ensemble uncertainty, and latent equivalence classes. The term *tomography* remains conditional if multiple hidden states explain the observations equally well.

6.8 Execution and result firewall

Draft 2 authorizes protocol construction, synthetic tests, and exact manifest preparation only. It does not authorize opening the final test identities, downloading outcomes for exploratory tuning, or writing a positive result in advance. The first empirical run must record environment, package versions, source URLs, file hashes, eligibility decisions, split hashes, random seeds, commands, and all failed or excluded runs. Any deviation from `benchmark-config.json` is timestamped and classified as preregistration correction or post hoc sensitivity analysis.

7. Draft 4-6 implementation tranches: synthetic contract, recurrent architecture, and action calibration

7.1 Scope and purpose

The first implementation tranche asks whether the frozen protocol can be executed without leakage and whether its analysis routes can recover a mechanism deliberately placed in synthetic data. This is a prerequisite for a biological benchmark, not a substitute for one. Passing means that the code can detect known structure and known destruction of that structure. It cannot show that a brain uses NAPOT.

The harness reads `protocol/benchmark-config.json` and refuses to run if the model registry is not exactly M0-M7, if the null registry does not contain eleven entries, or if the frozen protocol status no longer says that no data have been downloaded. It writes a deterministic split manifest and a machine-readable result file. The same command generated identical result bytes on two consecutive executions.

7.2 Synthetic system

Each of five frozen random seeds generated 15 synthetic mouse identities with 96 observations per mouse, for 1,440 rows per seed. Mouse identities were assigned by SHA-256 under the Draft 2 salt to nine development, three validation, and three test mice. No mouse appeared in more than one partition.

The generator contains a four-dimensional recurrent latent state plus stimulus, movement, rate, power, undirected connectivity, circular phase, three spatial coordinates, directed phase, dynamic gating, action history, and refferent observations. Refference contains an independent feedback innovation so that a model lacking that channel cannot reconstruct it perfectly from the latent state. A separate outcome-derived leakage sentinel is generated and then forbidden from every model registry. Future-neural and binary-action outcomes depend on a declared conjunction of spatial direction, gating, action memory, and refference, so the fixture has a known answer.

7.3 Capacity-matched proxy execution

This tranche is an interface and failure-detection test, not the final neural architecture. Every model admits only the feature blocks frozen for M0-M7. Development-only means and scales normalize those blocks. Raw standardized channels are retained, deterministic nonlinear features fill the remaining width, and every scalar prediction head receives exactly 32 mapped features plus an intercept. Ridge strength is selected on validation data from a fixed three-value grid. Test data do not select preprocessing, features, width, or regularization.

This equal-width proxy is deliberately weaker than the Draft 2 recurrent implementation. It establishes that all model contracts can run and that information restrictions are enforced. It does not satisfy the frozen

Adam, recurrent-dynamics, early-stopping, or full parameter-matching obligations for the eventual biological models.

7.4 Leakage and observability results

All five leakage audits passed:

1. development, validation, and test mouse sets were disjoint;
2. the outcome-derived sentinel was absent from all eight admitted feature sets;
3. perturbing test outcomes could not change fitted coefficients or selected regularization; and
4. normalization was fitted on development rows only.

The full observation set recovered the held-out synthetic latent state much better than rate alone:

Seed	Full-observation latent R-squared	Rate-only latent R-squared
20260718	0.9685	0.6291
20260719	0.9682	0.6153
20260720	0.9661	0.5382
20260721	0.9666	0.5965
20260722	0.9715	0.5938

This confirms observability of the generated state under the generated observation map. It does not establish observability in either Allen dataset.

7.5 Model-contract results

M7 beat the best M0-M6 proxy on both generated outcomes in every seed:

Seed	Neural MSE reduction versus best	
	M0-M6	Action log-loss reduction, bits
20260718	1.6761	0.0344
20260719	1.3081	0.0328
20260720	1.5738	0.0103
20260721	1.5398	0.0248
20260722	1.5454	0.0394
Mean	1.5287	0.0283

The result verifies power against a deliberately M7-like generator. It is not a confirmatory effect size, does not use bits per spike, and cannot be compared with the Draft 2 smallest effect sizes for biological data.

7.6 Null and ablation routing

Every frozen null and ablation route was exercised. In fixed-fit sensitivity checks, phase randomization, spatial coordinate permutation, direction disruption, temporal block shuffling, dynamic-gating ablation, reafference ablation, sensor dropout, and label permutation increased neural loss on average. Action-history permutation increased action log loss by 0.0091 bits on average. Movement removal produced little neural change and slightly improved synthetic action loss; this is reported rather than hidden because movement is a nuisance-control route, not a component the fixture must require.

These checks transform held-out features under a fixed fitted proxy. The biological protocol still requires each confirmatory null to follow its preregistered fitting and inference procedure. A fixed-fit sensitivity loss is evidence that the harness uses a channel, not proof that a refitted comparator cannot recover the outcome.

7.7 Executable causal phase and PWD operators

The second Draft 3 tranche implements the signal-processing contract in Equations (8)-(10) without opening biological data. For a configured set of center frequencies, the module uses a one-sided complex demodulation and exponentially weighted state update. The complex state at sample t therefore depends only on samples at or before t . Directed edges are represented as ordered sender-receiver pairs. Their instantaneous relative phase is measured at the receiver, while the phase and log-amplitude references used at t are computed only from observations through $t-1$. The resulting feature vector contains wrapped phase departure, log-amplitude departure, the causal run length of a thresholded departure, and a coordinate-normalized lagged phase gradient. That last quantity is an operational propagation feature, not a physical velocity estimate.

The validation fixture contained 2,400 samples, three channels, two directed edges, and the 9, 28, and 60 Hz measurement coordinates. It passed eight declared checks. Most importantly, replacing every input after sample 1,500 with large random perturbations changed no filter, baseline, PWD, duration, or lagged-gradient output at or before sample 1,500: the maximum difference was exactly zero. Perturbing the current receiver sample did not alter either baseline used for that sample, confirming that the reference was strictly past-only. Wrapped phase stayed in $[-\pi, \pi)$, reversing an edge reversed receiver-relative phase to numerical precision, event-duration counters followed their causal update rule, output shapes matched the declared contract, and all post-warmup outputs were finite.

This closes a software-mechanics subgate, not the biological-preprocessing gate. The module has not yet consumed Allen LFP, spike, coordinate, movement, pupil, or artifact schemas; its time constants, frequency coordinates, event threshold, and lag remain frozen protocol inputs rather than biological findings. It also does not implement the recurrent M0-M7 architectures or show that a PWD adds held-out information beyond rate and power.

7.8 Capacity-matched recurrent architecture tranche

Draft 4 replaces the equal-width ridge proxy with an executable recurrent suite while preserving the frozen M0-M7 information-admission order. Every model receives a 32-channel padded input, a 32-dimensional recurrent state, and separate neural and action heads under the same 2,146-trainable-parameter contract. Development-only normalization, validation-selected regularization, mouse-isolated 9/3/3 partitions, and exclusion of the outcome-derived leakage sentinel remain enforced. The suite uses the same five frozen seeds as the proxy gate.

The deliberately M7-like synthetic future-neural target produced a stable architecture-level recovery result:

Seed	M7 neural MSE	M7 minus best M0-M6 neural MSE	M7 action log loss, bits	M7 minus best M0-M6 action loss, bits
20260718	2.8731	-0.7787	0.9853	-0.00033
20260719	2.0742	-0.2625	0.9462	+0.02001
20260720	3.2329	-0.7384	0.8634	-0.00809
20260721	2.5225	-0.5011	1.0212	+0.00476
20260722	1.6193	-1.3889	0.9837	-0.01995
Median	2.5225	-0.7384	0.9837	-0.00033

M7 had the lowest future-neural test loss in all five seeds. This closes the previously open synthetic recovery-through-recurrent-architectures gate for the generated future-neural target. It shows that the capacity-matching and recurrence code can preserve a known joined dependency after replacing the proxy heads.

The action endpoint did not produce the same stability. M7 had the lowest action loss in three seeds, while M6 was better in two. The median difference was close to zero. Draft 4 therefore records the action result as inconclusive rather than as a pass. This distinction matters because NAPOT’s receive-transform-project-re-enter account predicts both state reconstruction and action-relevant consequence; a synthetic neural win cannot stand in for the second endpoint.

The architecture suite remains a generated-data test. Its positive future-neural result follows from a fixture intentionally containing the M7 conjunction, and its inconclusive action result exposes a sensitivity that must be addressed before biological interpretation. Neither result estimates biological effect size, validates the choice of recurrent family, or authorizes opening an Allen test set.

7.9 Independent-seed action sensitivity

Draft 6 asks why the original action result was unstable instead of hiding the instability behind the future-neural result. The generated action target was parameterized along two declared axes: the strength of the M7-distinctive directed-gating, action-history, and reafference terms, and the standard deviation of additional action-logit noise. Default arguments reproduce every frozen five-seed action and neural delta exactly. A separate set of 20 seeds then evaluates six signal strengths at baseline noise and four noise levels at baseline signal, with the shared baseline counted once. This produces nine conditions and 180 capacity-matched M0-M7 comparisons.

The delta below is M7 action log loss minus the best M0-M6 action log loss; negative values favor M7. The descriptive stability rule was fixed as at least 16 wins in 20 seeds and a negative 90th percentile delta. This is a calibration rule for generated data, not a biological significance test.

Action condition	Wins / 20	Median delta, bits	10th percentile	90th percentile	Stability rule
signal 0.00, noise 0.35	6	+0.00045	-0.00169	+0.00948	fail
signal 0.25, noise 0.35	14	-0.00092	-0.00756	+0.00654	fail
signal 0.50, noise 0.35	14	-0.00422	-0.01675	+0.00337	fail
signal 0.75, noise 0.35	15	-0.00662	-0.01523	+0.00692	fail
signal 1.00, noise 0.35	15	-0.00570	-0.01946	+0.00585	fail
signal 1.50, noise 0.35	15	-0.00741	-0.01954	+0.00767	fail
signal 1.00, noise 0.00	15	-0.00527	-0.02213	+0.00735	fail
signal 1.00, noise 0.75	16	-0.00595	-0.01652	+0.00415	fail
signal 1.00, noise 1.25	15	-0.00776	-0.01835	+0.00309	fail

The zero-signal negative control behaves as required: adding the M7 feature set does not create a stable advantage when the generated target omits its distinctive terms. The baseline condition shows a modest directional tendency, but not a stable separation: five seeds favor a comparator, and the upper tail crosses zero. Increasing signal strength or changing noise does not repair that failure. None of the nine conditions passes the declared rule.

This result prevents two overclaims. First, the original three-of-five result cannot be promoted merely by increasing the seed count; the 20-seed baseline remains unstable. Second, the failure is not simply evidence that the synthetic action signal was too weak, because a 1.5-fold signal does not stabilize the advantage. The current action head, recurrent state, comparator family, or binary endpoint may be inadequately specified. Draft 6 therefore retains the action gate as open and requires redesign to be justified on development data before any biological test boundary is opened. This synthetic calibration does not estimate a biological effect size.

7.10 Final endpoint and future biological test

This paper intentionally ends before biological execution. The following items define the separately governed Allen study that would test the frozen protocol; they are not unfinished requirements for the present

genealogy, protocol, and synthetic failure-calibration paper:

- populate exact Allen file manifests with URLs, byte counts, SHA-256 hashes, session identities, and eligibility decisions;
- create biological mouse and within-development split manifests without opening test outcomes;
- bind the validated causal phase / PWD operators to real Allen LFP schemas, then implement artifact masks and movement / eye covariates without test access;
- execute every comparator, null, ablation, calibration, sensitivity, and uncertainty analysis;
- report every positive, null, contradictory, and failed result; and
- obtain independent code, leakage, neuroscience, and statistical review.

The template manifests/allen-file-manifest.template.json keeps that future-study gate closed: its file arrays are empty and biological test identities and outcomes remain unopened. This separation prevents an unexecuted biological study from being mistaken for a hidden positive result, while allowing the present negative calibration and frozen methods to stand as complete reportable results. # 8. Falsifiable predictions

The current framework makes the following bounded predictions.

Prediction 1: directional phase increment. A model containing measured phase gradients and propagation direction should improve a preregistered held-out outcome beyond local rate and power when the dataset resolves those variables. Phase randomization preserving power should remove that increment.

Prediction 2: dynamic subnetwork increment. Context-dependent gating should predict trial-to-trial changes in effective pathways better than one fixed connectivity matrix. If a fixed model generalizes equally well, revision 4’s gating claim is unnecessary for that task.

Prediction 3: inceptive-field increment. A model trained to predict action and reafferent sensory consequence should generalize across selected interventions better than a receptive-field-only model. If it predicts only correlated action in familiar trials, the stronger action-side claim fails.

Prediction 4: multiscale increment. Nested time scales should improve cross-context prediction beyond the best single-scale model. Fixed frequency labels are not part of the prediction; the relevant scales must be selected within the training data and frozen before testing.

Prediction 5: selective integration. Perturbing a component identified as part of the reconstructed state should cause a specific downstream error and recovery trajectory. Global synchrony without selective causal consequences does not support entification.

Prediction 6: reconstruction-to-action continuity. Latent variables that best reconstruct the relevant body / world state should also contribute to action and consequence prediction. A geometry that decodes stimuli but has no incremental relation to behavior is insufficient for the proposed sensorimotor rendering role.

9. Failure conditions

NAPOT should be rejected or narrowed for a dataset if:

1. the full model does not beat capacity-matched flexible baselines;
2. apparent gains disappear after movement, arousal, or leakage controls;
3. phase gains survive phase destruction, indicating an artifact;
4. spatial direction can be reversed without loss;
5. inferred latent state does not generalize across the frozen boundary;
6. action and reafference predictions add no value;
7. observability or simulation recovery shows that the claimed state cannot be identified; or

8. perturbations contradict the inferred causal routing.

A negative result is informative. It can show that a historical revision is unnecessary, that the measurement resolution is inadequate, or that the current operationalization is wrong. It should not be converted into an untestable claim that the true rendering occurs elsewhere.

10. Relationship to established work

Intrinsic neuronal oscillation and resonance are established electrophysiological subjects. Llinas distinguished neurons whose conductances support autorhythmic pacemaking from neurons that respond preferentially to particular input frequencies [13]. That result supports treating neural timing selectivity as a measurable cellular property. It does not establish NAPOT’s multiscale reconstruction, directed projection, action closure, tomography, or consciousness claims.

Propagating cortical activity, feedforward and feedback interactions, and low-dimensional neural structure are also established research subjects. Muller and colleagues reported stimulus-evoked propagating population responses in awake monkey visual cortex. Aggarwal and colleagues reported oppositely directed fast feedforward and slow feedback visual-evoked waves in awake mice. Cunningham and Yu reviewed dimensionality-reduction methods for large-scale neural recordings. These works establish measurements and comparator questions. They do not validate NAPOT.

Close theory and computational comparators further divide the proposal into parts that can be tested separately. Hawkins and Ahmad model active dendritic segments as branch-sensitive contextual inputs for sequence memory [8]. The Thousand Brains framework proposes location-based reference frames and distributed object models across cortical columns [9], and Lewis and colleagues implement a sensorimotor object-recognition model using cortical location signals [10]. George and colleagues show that clone-structured cognitive graphs can disambiguate aliased observations while learning latent spatial or sequential structure [11]. These works make dendritic context, coordinate-aware sensorimotor inference, and latent-state disambiguation serious comparators. They do not establish NAPOT’s directed phase features, receive-transform-project loop, action / refference constraint, neural tomography, or consciousness claims.

The benchmark therefore cannot ask only whether NAPOT decodes a stimulus. It must ask whether oscillatory direction and dynamic routing add held-out value beyond active-dendrite, coordinate, graph, and flexible state-space alternatives; whether that value survives movement and phase-destruction controls; and whether the reconstructed variables predict action and sensory consequence. The claim-by-claim source boundary is recorded in reviews/DRAFT3-COMPARATOR-SOURCE-AUDIT-20260719.md.

NAPOT’s proposed delta is the joined constraint set:

```
partial distributed observations
+ branch- and state-sensitive transformation
+ directional multiscale temporal structure
+ dynamic effective subnetworks
+ recurrent sensory-to-action consequence
+ intervention-tested latent reconstruction
```

Each element has relatives in ordinary neuroscience. The paper’s value depends on whether their conjunction yields a more accurate and more falsifiable model than the alternatives. A new acronym is not evidence of a new mechanism.

11. Consciousness and the inner-viewer boundary

This Draft 6 does not identify a reconstructed latent state with phenomenal consciousness. It makes a narrower contribution to consciousness research: it specifies how a distributed system might maintain an integrated, body-centered, action-relevant state without requiring an internal spectator. That closes one explanatory gap but leaves others open, including why a state is experienced, which states are reportable, and what distinguishes conscious from unconscious integration.

Likewise, a three-dimensional decoder does not prove that the brain contains a three-dimensional internal picture. A relational state can guide reach, avoidance, gaze, and prediction without being formatted as a viewed scene. Future consciousness claims require independent contrasts, perturbations, and comparisons with unconscious processing. They cannot be inferred from model fit alone.

12. Role in SAN Book 2

The paper can support Book 2 in three bounded ways. First, it preserves the actual ten-stage development instead of presenting NAPOT as a finished concept that appeared all at once. Second, it gives the reader a plain-language receive-transform-project-re-enter loop before the acronym. Third, it supplies the formal and empirical gates that keep rendering and tomography from becoming unfalsifiable metaphors.

The full ten-row source ledger belongs in the Evidence Atlas or appendix. Chapter 18 needs only the explanatory sequence: each revision increased the resolution of the question, connected sensory inference to action, or imposed a stronger test. NAPOT 7 should be marked as the historical physics fork and routed to the QGTCD paper rather than used as evidence for neural rendering.

13. Conclusion

NAPOT has advanced from a ten-stage genealogy, through a frozen model-comparison protocol, to executable synthetic model-contract, causal signal-operator, capacity-matched recurrent-architecture, and action-sensitivity gates. The genealogy explains why the candidate contains arrays, receiver-relative tonic and phasic roles, dynamic gating, directed phase, sensory reconstruction, action consequence, and intervention tests. The protocol defines how those components can fail. The implementation tranches now demonstrate that all eight restricted model routes and eleven destruction routes can execute under deterministic, mouse-isolated splits, that the declared phase / PWD features can be computed without future-sample leakage, and that the joined M7 information contract can survive an equal-parameter recurrent implementation on a generated future-neural target.

The synthetic fixture passed its declared observability, leakage, model-increment, and targeted-null checks across all five frozen seeds. The recurrent M7 model also beat every M0-M6 comparator on future-neural MSE in five of five seeds, but beat them on action loss in only three of five. The 180-comparison sensitivity study sharpened that limitation: the zero-signal control failed as intended, the baseline favored M7 in 15/20 independent seeds, and none of nine signal / noise conditions passed the upper-tail stability rule. This is engineering evidence about the benchmark harness and its failure behavior. It is not neuroscience evidence, a consciousness result, or validation of neural rendering. The fixture was designed to contain the conjunction M7 is meant to detect, and the unstable action endpoint remains an explicit warning against promoting one synthetic success into a joined-mechanism result.

No Allen outcome has been opened. This final-preprint endpoint closes live-video custody, the frozen biological protocol, executable causal operators, matched synthetic model routing, and the adverse

action-stability result. Exact biological manifests, schema-bound preprocessing, confirmatory null refits, uncertainty analyses, and complete outcome reporting belong to the separately governed Allen execution. A later positive benchmark would support only the tested components on the tested datasets. It would not establish an internal spectator, a literal neural screen, or the identity of a decoded manifold with consciousness. In this account the network itself remains the observer-action process: distributed cells receive differences, alter shared state, project consequences, and re-enter the loop.

Protocol, data, and code availability

The frozen protocol is `protocol/benchmark-config.json`. The synthetic model-contract harness is `implementation/run_synthetic_contract_benchmark.py`; it writes `results/draft3-synthetic-contract-results.json` and `manifests/draft3-synthetic-split-manifest.json`. The causal signal module and its validator are `implementation/causal_signal_operators.py` and `implementation/validate_causal_signal_operators.py`; the validator writes `results/draft3-causal-operator-validation.json`. The recurrent architecture suite is `implementation/run_draft3_recurrent_architecture_benchmark.py`; its machine-readable result and independent gate receipt are `results/draft3-recurrent-architecture-results.json` and `results/draft3-recurrent-architecture-validation.json`. The independent-seed calibration is `implementation/run_draft6_action_sensitivity.py`; its result and validator receipt are `results/draft6-action-sensitivity.json` and `results/draft6-action-sensitivity-validation.json`. The empty biological file contract is `manifests/allen-file-manifest.template.json`. The primary and external datasets remain available through official AllenSDK documentation and versioned manifests [5-7], but no biological file, test identity, or outcome was opened in this tranche. A future Allen execution requires exact downloaded-file hashes, biological split manifests, schema-bound preprocessing, confirmatory null refits, uncertainty outputs, and complete outcome manifests. The current synthetic package is bound to `ENVIRONMENT-LOCK.json`. The governed 2022 public-source ledgers, raw-transcript hashes, and metadata / archive receipt are listed in `SOURCE-MAP.md`; they are provenance inputs rather than model-training or biological-result data.

Authorial provenance and research chronology

Artificial intelligence was not involved in authoring or originating the core ideas, scientific theories, hypotheses, or source-based research presented in this paper. The underlying research program was not written or developed by AI. This body of work is Micah Blumberg’s original research, developed through a continuous program spanning 2011-2026. Its central conceptual foundations were developed and recorded before ChatGPT’s public release on November 30, 2022.

In later manuscript development, AI tools were used only in bounded supporting roles, including assistance with mathematical and computational formulation, code, grammar and spelling, and checks against medical, biological, and physics literature. These tools did not supply the original theories or authorship. Micah Blumberg selected the claims and sources, directed the analysis, approved the final language, and remains responsible for the paper’s arguments, interpretations, and errors.

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